

ASSIGNMENT No. 01 – Data Preprocessing on Employee Dataset using WEKA

Subject:	Data Mining using WEKA	Tool:	WEKA Explorer
Dataset:	Employee.csv (10 instances, 8 attributes)	Total Qs:	20

■ Employee.csv – Dataset

Create the following CSV file and save as **Employee.csv**:

EmpID	Ename	Department	Gender	Age	Salary	Experience	Performance
201	Amit	IT	Male	24	35000	2	High
202	Priya	HR	Female	27	40000	3	Medium
203	Ramesh	Sales	Male	30	?	5	Low
204	Sneha	IT	Female	26	45000	?	High
205	Kunal	Finance	Male	29	48000	4	Medium
206	Meena	HR	Female	28	42000	3	High
207	Rajesh	IT	Male	31	50000	6	Low
208	Shital	Sales	Female	25	36000	2	?
209	Vinay	Finance	Male	?	55000	7	Medium
210	Kavita	HR	Female	23	38000	1	High

Note: Cells with ? represent missing values. In WEKA, '?' is the standard placeholder for missing data.

Q1. Load the Dataset into WEKA

Procedure:

Open WEKA → Click on **Explorer**

In the **Preprocess** tab → Click 'Open file...'

Change file type filter to **CSV files (*.csv)**

Browse to **Employee.csv** → Click **Open**

WEKA loads the dataset: displays **10 instances** and **8 attributes**

Verify attribute types: EmpID (Numeric), Ename (String), Department (Nominal), Gender (Nominal), Age (Numeric), Salary (Numeric), Experience (Numeric), Performance (Nominal)

To save as ARFF for future use: **Save** → **Employee.arff**

Q2. Add New Attribute 'Grade' (Nominal: {A, B, C}) at Index 6

Procedure:

Preprocess tab → Click '**Choose**' under Filters section

Navigate to: **unsupervised** → **attribute** → **Add**

Click on the filter name to open configuration

Set **attributeIndex = 6**

Set **attributeName = Grade**

Set **attributeType = NOM** (Nominal)

Set **nominalLabels = A,B,C**

Click **OK** → then click **Apply**

Verify: 'Grade' attribute appears at position 6 with values {A, B, C}

Index	Attribute Name	Type	Values/Range
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1	EmpID	Numeric	201–210
2	Ename	String/Nominal	Names
3	Department	Nominal	IT, HR, Sales, Finance
4	Gender	Nominal	Male, Female
5	Age	Numeric	23–31
6	Grade	Nominal (NEW)	A, B, C
7	Salary	Numeric	35000–55000
8	Experience	Numeric	1–7
9	Performance	Nominal	High, Medium, Low

Q3. List All Attributes in the Dataset

Procedure:

In the **Preprocess** tab, the **Attributes** panel (left side) lists all attributes

Click on each attribute name to view its statistics (min, max, mean, missing count)

#	Attribute	Type	Description	Missing?
1	EmpID	Numeric	Employee ID (identifier)	No
2	Ename	Nominal/String	Employee Name	No
3	Department	Nominal	IT / HR / Sales / Finance	No
4	Gender	Nominal	Male / Female	No
5	Age	Numeric	Employee Age in years	Yes (1 — Vinay)
6	Salary	Numeric	Monthly Salary in Rs.	Yes (1 — Ramesh)
7	Experience	Numeric	Years of experience	Yes (1 — Sneha)
8	Performance	Nominal	High / Medium / Low	Yes (1 — Shital)

Q4. Identify Attributes with Missing Values and Calculate Percentage

Procedure:

In Preprocess tab, click each attribute → check **Missing** count shown on right panel

$\text{Missing \%} = (\text{Missing Count} / \text{Total Instances}) \times 100 = (n / 10) \times 100$

Attribute	Total Instances	Missing Count	Missing %	Status
EmpID	10	0	0%	Complete
Ename	10	0	0%	Complete
Department	10	0	0%	Complete
Gender	10	0	0%	Complete
Age	10	1 (Vinay-row 9)	10%	Low missing
Salary	10	1 (Ramesh-row 3)	10%	Low missing
Experience	10	1 (Sneha-row 4)	10%	Low missing
Performance	10	1 (Shital-row 8)	10%	Low missing

Result: 4 attributes have missing values — Age, Salary, Experience, Performance — each with 10% missing (1 out of 10 instances).

Q5. Remove Attribute if Missing Values > 40%

Analysis:

Threshold: 40% of 10 = 4 instances

All attributes have only 10% missing (1 instance each) — well below 40%

Result: NO attribute is removed — no attribute exceeds the 40% threshold.

If removal was needed, the procedure would be:

Preprocess → Filter → **unsupervised** → **attribute** → **Remove**

Set **attributeIndices** to the index of the attribute to remove → Click Apply

Alternative: **unsupervised** → **attribute** → **RemoveUseless** (auto-detects useless attrs)

Q6. Convert EmpID Attribute from Numeric to Nominal

Procedure:

Filter → Choose → **unsupervised** → **attribute** → **NumericToNominal**

Click on filter name → set **attributeIndices = 1** (EmpID is at index 1)

Click **OK** → then **Apply**

Verify: EmpID attribute type changes from **Numeric** to **Nominal**

Why? EmpID is an identifier — arithmetic on it (mean, std) is meaningless. Converting to Nominal prevents WEKA from treating it as a measurable quantity.

Q7. Replace Missing Values (ReplaceMissingValues Filter)

Procedure:

Filter → Choose → **unsupervised** → **attribute** → **ReplaceMissingValues**

No additional configuration needed (applies to all attributes by default)

Click **Apply**

WEKA replaces missing values using:

- **Numeric attributes** → replaced with the **Mean** of that attribute
- **Nominal attributes** → replaced with the **Mode** (most frequent value)

Attribute	Type	Missing Row	Replacement Method	Replaced Value
Age	Numeric	Vinay (row 9)	Mean of Age	Mean = $(24+27+30+26+29+28+31+25+23)/9 = 27$
Salary	Numeric	Ramesh (row 3)	Mean of Salary	Mean = $(35000+40000+45000+48000+42000+50000+36000+55000+38000)/9 = 43222$
Experience	Numeric	Sneha (row 4)	Mean of Experience	Mean = $(2+3+5+4+3+6+2+7+1)/9 \approx 3.67 \approx 4$
Performance	Nominal	Shital (row 8)	Mode	Mode = High (appears 4 times)

Q8. Discretize Salary into 3 Bins (Low, Medium, High)

Procedure:

Filter → Choose → **unsupervised** → **attribute** → **Discretize**

Click on filter name → Set **attributeIndices = 6** (Salary after Grade added at 6, or original index 6 = Salary)

Set **bins = 3**

Set **useEqualFrequency = False** (equal-width bins)

Click OK → Apply

Salary Range: Min=35000, Max=55000 | Width per bin = $(55000-35000)/3 = 6667$

Bin	Range	Label	Employees
Bin 1 (Low)	35000 – 41667	Low	Amit(35000), Priya(40000), Kavita(38000), Shital(36000)
Bin 2 (Medium)	41667 – 48333	Medium	Ramesh(?≈43222), Sneha(45000), Meena(42000), Kunal(48000)
Bin 3 (High)	48333 – 55000	High	Rajesh(50000), Vinay(55000)

Note: WEKA labels bins as '(-inf-41667]', '(41667-48333]', '(48333-inf)'. You can rename these manually in the ARFF editor.

Q9. Copy the Experience Attribute to the 4th Position

Procedure:

Filter → Choose → **unsupervised** → **attribute** → **Copy**

Set **attributeIndices = 7** (Experience is at index 7 after Grade added)

Click Apply → A copy of Experience is appended at the end
 Then use **Reorder** filter to place the copy at position 4:
 Filter → **unsupervised** → **attribute** → **Reorder**
 Enter indices: **1,2,3,10,4,5,6,7,8,9** (placing copied Experience at position 4)
 Click OK → Apply

Result: Dataset now has Experience copied at the 4th column position.

Q10. Detect and Remove Outliers in the Age Attribute

Procedure – Detection using InterquartileRange filter:

Filter → **unsupervised** → **attribute** → **InterquartileRange**
 Set **attributeIndices = 5** (Age)
 Set **outlierFactor = 3.0**, **extremeValuesFactor = 6.0**
 Click Apply → WEKA adds 'Outlier' and 'ExtremeValue' flag attributes

Manual IQR Calculation for Age:

Sorted Ages (known): 23, 24, 25, 26, 27, 28, 29, 30, 31 (Vinay's age is missing)
 $Q1 = 25$, $Q3 = 29$, $IQR = Q3 - Q1 = 4$
 $Lower\ Bound = Q1 - 1.5 \times IQR = 25 - 6 = 19$
 $Upper\ Bound = Q3 + 1.5 \times IQR = 29 + 6 = 35$

All Ages (23–31) fall within [19, 35] → No outliers detected

Removal Procedure (if outliers existed):

Filter → **unsupervised** → **instance** → **RemoveWithValues**
 Set **attributeIndex** to 'Outlier' column → Set **nominalIndices = 1** → Apply

Q11. Normalize the Salary Attribute using Normalize Filter

Procedure:

Filter → **unsupervised** → **attribute** → **Normalize**
 Set **attributeIndices = 6** (Salary column)
 Set **scale = 1.0**, **translation = 0.0** (default → scales to [0,1])
 Click OK → Apply
Formula: $Normalized = (x - min) / (max - min) = (x - 35000) / (55000 - 35000)$

EmpID	Name	Original Salary	Normalized Value	Calculation
201	Amit	35000	0.000	$(35000-35000)/20000$
202	Priya	40000	0.250	$(40000-35000)/20000$
203	Ramesh	? → 43222	0.411	$(43222-35000)/20000$
204	Sneha	45000	0.500	$(45000-35000)/20000$
205	Kunal	48000	0.650	$(48000-35000)/20000$
206	Meena	42000	0.350	$(42000-35000)/20000$
207	Rajesh	50000	0.750	$(50000-35000)/20000$
208	Shital	36000	0.050	$(36000-35000)/20000$
209	Vinay	55000	1.000	$(55000-35000)/20000$
210	Kavita	38000	0.150	$(38000-35000)/20000$

Q12. Standardize the Age Attribute

Procedure:

Filter → **unsupervised** → **attribute** → **Standardize**
 Set **attributeIndices = 5** (Age column)
 Click OK → Apply → Age values are converted to Z-scores
Formula: $Z = (x - mean) / std_dev$

Known ages: 24,27,30,26,29,28,31,25,23 (9 values, excluding Vinay's missing)

Mean = $(24+27+30+26+29+28+31+25+23)/9 = 243/9 = 27$

Variance = $[(24-27)^2+(27-27)^2+(30-27)^2+(26-27)^2+(29-27)^2+(28-27)^2+(31-27)^2+(25-27)^2+(23-27)^2] / 9$
 $= [9+0+9+1+4+1+16+4+16] / 9 = 60/9 = 6.67 \rightarrow \text{Std Dev} = \sqrt{6.67} \approx 2.58$

EmpID	Name	Age	Z-score = $(\text{Age}-27)/2.58$
201	Amit	24	$(24-27)/2.58 = -1.163$
202	Priya	27	$(27-27)/2.58 = 0.000$
203	Ramesh	30	$(30-27)/2.58 = +1.163$
204	Sneha	26	$(26-27)/2.58 = -0.388$
205	Kunal	29	$(29-27)/2.58 = +0.775$
206	Meena	28	$(28-27)/2.58 = +0.388$
207	Rajesh	31	$(31-27)/2.58 = +1.550$
208	Shital	25	$(25-27)/2.58 = -0.775$
209	Vinay	? $\rightarrow 27$	$(27-27)/2.58 = 0.000$ (replaced by mean)
210	Kavita	23	$(23-27)/2.58 = -1.550$

Q13. Remove Duplicate Records

Procedure:

Filter $\rightarrow$ Choose $\rightarrow$ **unsupervised** $\rightarrow$ **instance** $\rightarrow$ **RemoveDuplicates**

No additional configuration required

Click **Apply**

WEKA scans all instances and removes exact duplicate rows

Result: In the Employee dataset, all 10 EmpIDs (201–210) are unique. Therefore, no duplicate records exist — all 10 instances are retained.

Q14. Apply Attribute Selection and List Best Subset of Features

Procedure (using Select Attributes tab):

Go to the **Select Attributes** tab in WEKA Explorer

Attribute Evaluator: Choose **CfsSubsetEval** (Correlation-based Feature Selection)

Search Method: Choose **BestFirst** or **GreedyStepwise**

Set class attribute to **Performance** (last attribute)

Click **Start** $\rightarrow$ WEKA evaluates all feature subsets

Alternative – Filter method:

Preprocess $\rightarrow$ Filter $\rightarrow$ **supervised** $\rightarrow$ **attribute** $\rightarrow$ **AttributeSelection**

Evaluator: **InfoGainAttributeEval**, Ranker method

Rank	Attribute	Info Gain (Estimated)	Selected?
1	Salary	High (0.40+)	✓ Yes
2	Experience	High (0.35+)	✓ Yes
3	Department	Medium (0.25+)	✓ Yes
4	Age	Medium (0.20+)	✓ Yes
5	Gender	Low (0.10)	Optional
6	EmpID	None (identifier)	✗ Remove
7	Ename	None (identifier)	✗ Remove

Best Subset: {Salary, Experience, Department, Age} for predicting Performance.

Q15. Replace Missing Numeric Values with 0 and Nominal with 'Unknown'

Method 1 – Edit CSV directly:

Open Employee.csv in a text editor

Replace all '?' in numeric columns (Age, Salary, Experience) with **0**

Replace '?' in nominal column (Performance) with **Unknown**

Save and reload in WEKA

Method 2 – Using WEKA filters:

For numeric: Filter → **unsupervised** → **attribute** → **ReplaceMissingWithUserDefinedValues**

Set numericReplacementValue = **0**

Set nominalReplacementValue = **Unknown**

Click Apply

Attribute	Type	Original Missing	Replaced With
Age	Numeric	? (Vinay)	0
Salary	Numeric	? (Ramesh)	0
Experience	Numeric	? (Sneha)	0
Performance	Nominal	? (Shital)	Unknown

Q16. Retrieve All Instances where Salary > 45000

Procedure:

Filter → Choose → **unsupervised** → **instance** → **RemoveWithValues**

Set **attributeIndex** = **6** (Salary)

Set **splitPoint** = **45000**

Set **invertSelection** = **True** (to KEEP rows with Salary > 45000)

Click OK → Apply

EmpID	Ename	Department	Salary	Condition
205	Kunal	Finance	48000	48000 > 45000 ✓
207	Rajesh	IT	50000	50000 > 45000 ✓
209	Vinay	Finance	55000	55000 > 45000 ✓

Result: 3 instances retrieved — Kunal (48000), Rajesh (50000), Vinay (55000)

Note: Ramesh's salary is missing ('?'). After replacement with mean (~43222), he would NOT qualify. Sneha (45000) also does not qualify (not strictly greater than).

Q17. Remove 25% of Instances using RemovePercentage Filter

Procedure:

Filter → Choose → **unsupervised** → **instance** → **RemovePercentage**

Set **percentage** = **25**

Set **invertSelection** = **False** (remove 25%, keep 75%)

Click OK → Apply

Calculation:

Total instances = 10

25% of 10 = 2.5 ≈ 2 or 3 instances removed (WEKA rounds)

Remaining instances = 7 or 8 (WEKA removes the last 25% by default)

Result: Approximately 2–3 instances (last rows) are removed. Remaining ~7–8 instances form the reduced dataset.

Tip: Use Randomize filter first to shuffle data before removing percentage so that removal is random rather than always from the end.

Q18. Produce a 20% Random Resample of the Dataset

Procedure:

Filter → Choose → **unsupervised** → **instance** → **Resample**

Set **sampleSizePercent = 20**

Set **randomSeed = 1** (for reproducibility)

Set **noReplacement = True** (sampling without replacement)

Click OK → Apply

Calculation:

20% of 10 instances = 2 instances will be in the resample

Result: A new dataset containing 2 randomly selected instances from the original 10.

With replacement (noReplacement=False): The same instance can appear multiple times. This is useful for bootstrapping. Without replacement, each instance appears at most once.

Q19. Change Attribute Name: Ename → Employee_Name

Procedure:

Filter → Choose → **unsupervised** → **attribute** → **RenameAttribute**

Set **attributeIndex = 2** (Ename is the 2nd attribute)

Set **newName = Employee_Name**

Click OK → then click **Apply**

Verify: In the Attributes panel, 'Ename' is now renamed to 'Employee_Name'

Alternative method: Open the ARFF file in a text editor → Find '@attribute Ename' → Change to '@attribute Employee_Name' → Save → Reload in WEKA

Q20. Save Processed Dataset as Employee_processed.arff

Procedure:

After applying all preprocessing filters in the Preprocess tab

Click the '**Save...**' button in the Preprocess tab

In the save dialog, change the filename to **Employee_processed.arff**

Make sure file format is set to **ARFF (Attribute-Relation File Format)**

Click **Save**

Sample ARFF File Structure:

ARFF File Content for Employee_processed.arff	
@relation	Employee_processed
@attribute	EmpID {201,202,203,204,205,206,207,208,209,210}
@attribute	Employee_Name {Amit,Priya,Ramesh,Sneha,Kunal,Meena,Rajesh,Shital,Vinay,Kavita}
@attribute	Department {IT,HR,Sales,Finance}
@attribute	Gender {Male,Female}
@attribute	Age numeric
@attribute	Grade {A,B,C}
@attribute	Salary numeric
@attribute	Experience numeric
@attribute	Performance {High,Medium,Low}
@data	
201,Amit,IT,Male,24,?,35000,2,High	
202,Priya,HR,Female,27,?,40000,3,Medium	
... (remaining instances) ...	

■ QUICK REFERENCE SUMMARY – All 20 Steps

Q#	Task	WEKA Filter / Method	Key Setting
1	Load Dataset	Explorer → Preprocess → Open file	Select Employee.csv
2	Add Grade attribute	unsupervised → attribute → Add	index=6, type=NOM, labels=A,B,C
3	List attributes	Preprocess → Attributes panel	Click each attribute
4	Find missing values	Click attribute → check Missing count	10% missing for 4 attrs
5	Remove >40% missing	unsupervised → attribute → Remove	None removed (all have 10%)
6	EmpID to nominal	unsupervised → attribute → NumericToNominal	attributeIndices=1
7	Replace missing values	unsupervised → attribute → ReplaceMissingValues	Default (mean/mode)
8	Salary into 3 bins	unsupervised → attribute → Discretize	bins=3, index=6
9	Copy Experience to pos 4	unsupervised → attribute → Copy + Reorder	Reorder indices
10	Detect/Remove outliers	unsupervised → attribute → InterquartileRange	No outliers found
11	Normalize Salary	unsupervised → attribute → Normalize	scale=1.0, index=6
12	Standardize Age	unsupervised → attribute → Standardize	index=5
13	Remove duplicates	unsupervised → instance → RemoveDuplicates	No duplicates found
14	Attribute selection	Select Attributes tab → CfsSubsetEval	Best: Salary,Exp,Dept,Age
15	Replace with constant	ReplaceWithMissingValues or edit CSV	Numeric→0, Nominal→Unknown
16	Salary > 45000	unsupervised → instance → RemoveWithValues	splitPoint=45000, invert=True
17	Remove 25% instances	unsupervised → instance → RemovePercentage	percentage=25
18	20% random resample	unsupervised → instance → Resample	sampleSizePercent=20
19	Rename Ename	unsupervised → attribute → RenameAttribute	newName=Employee_Name
20	Save as ARFF	Preprocess → Save → Employee_processed.arff	ARFF format